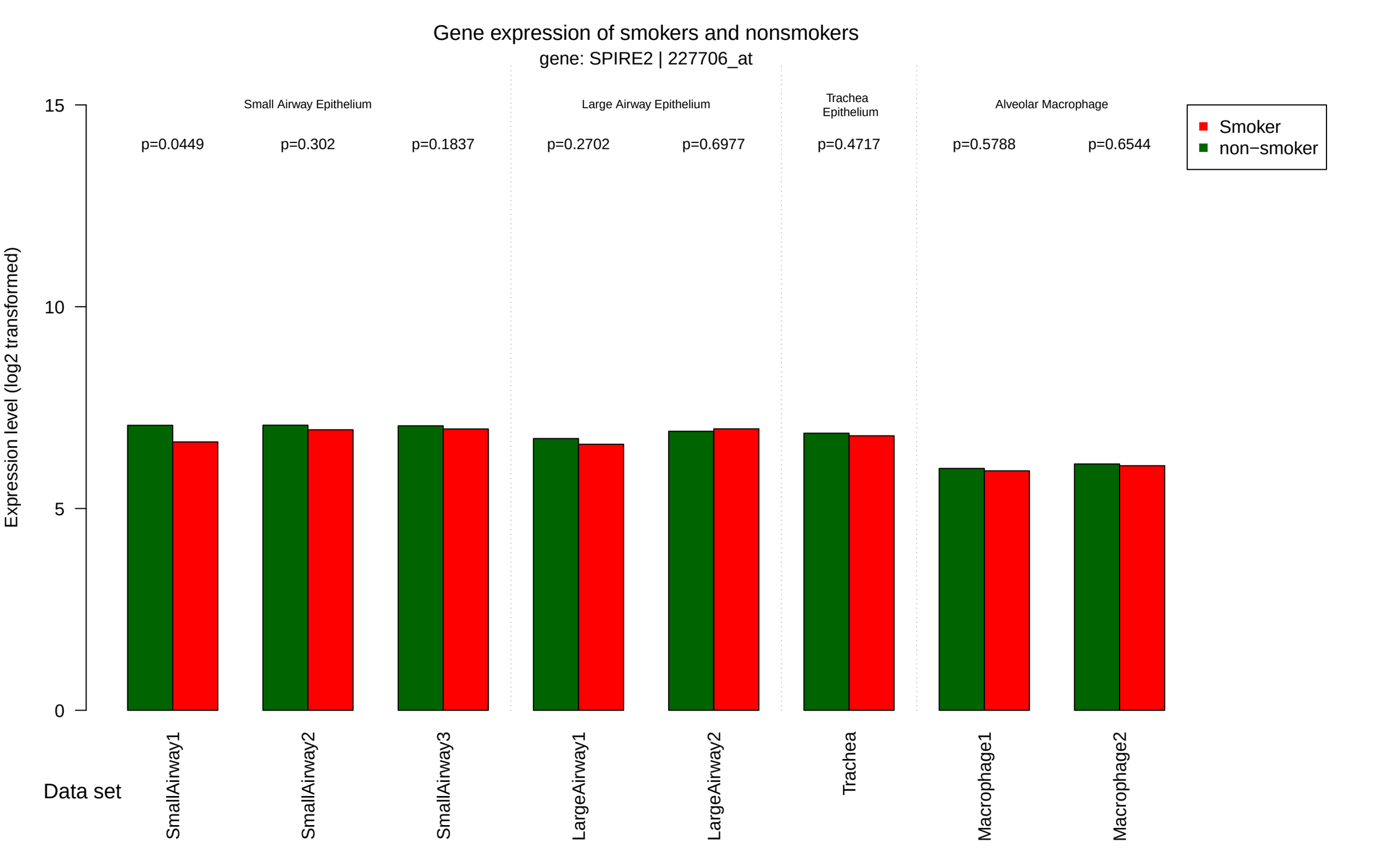
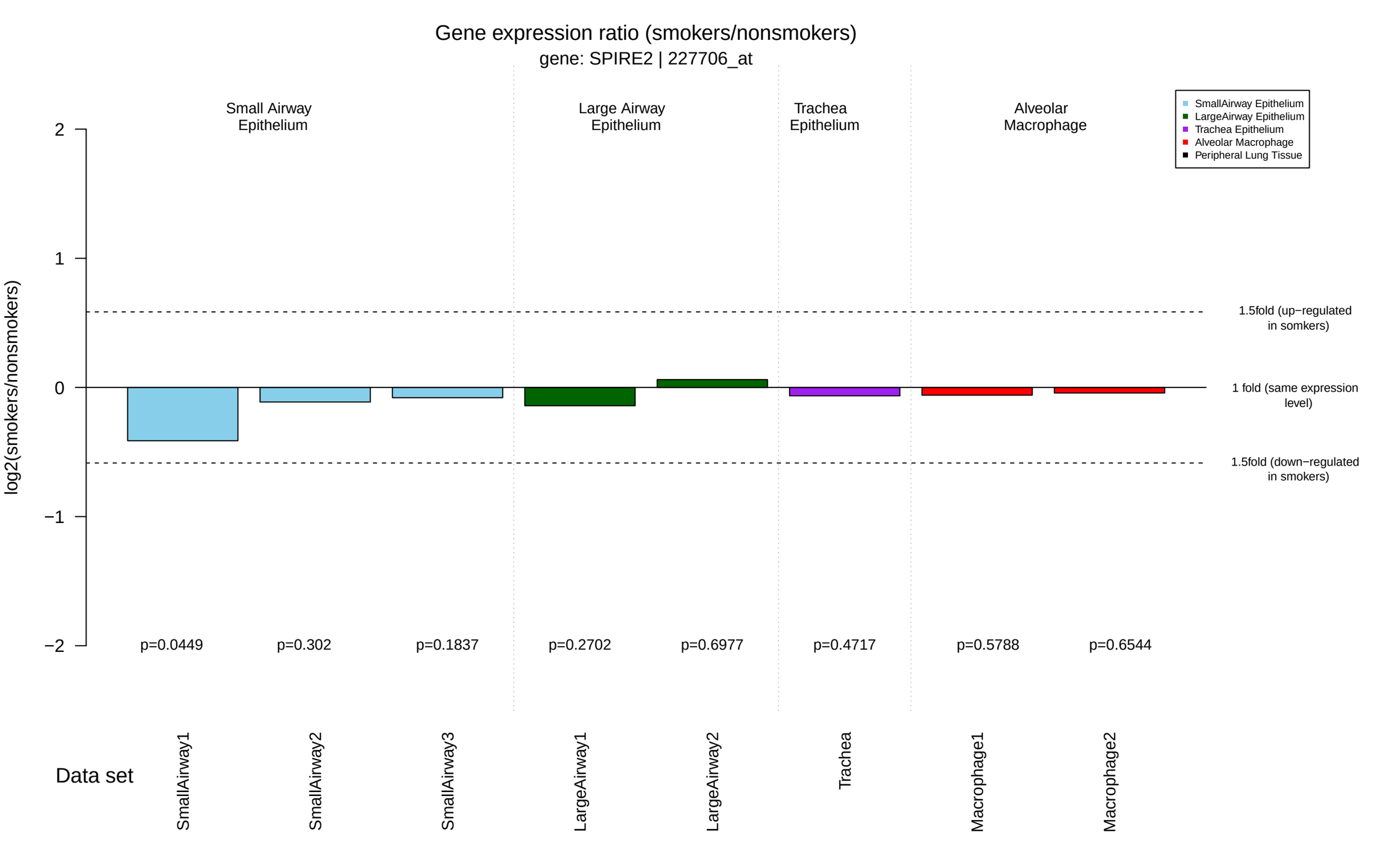
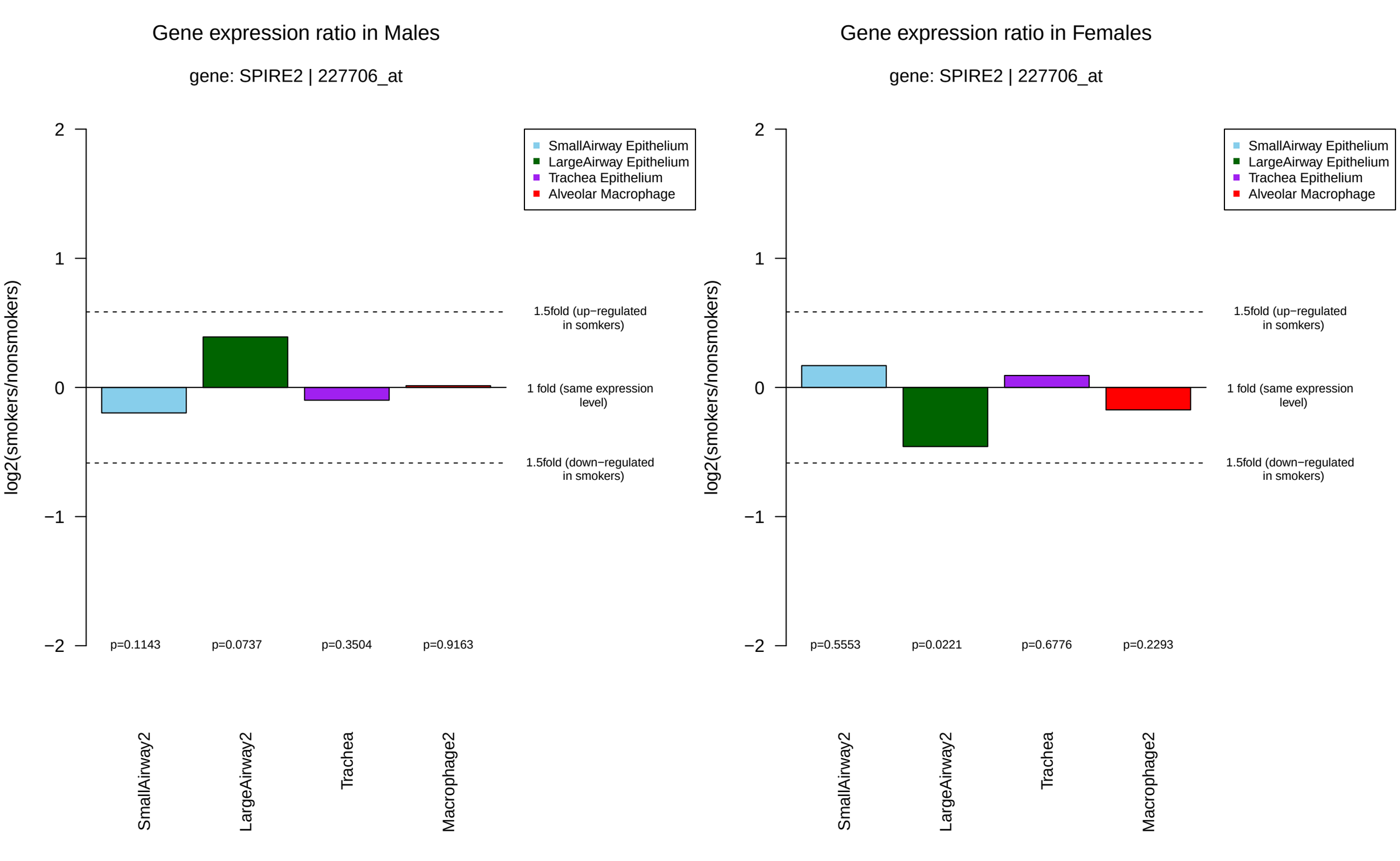
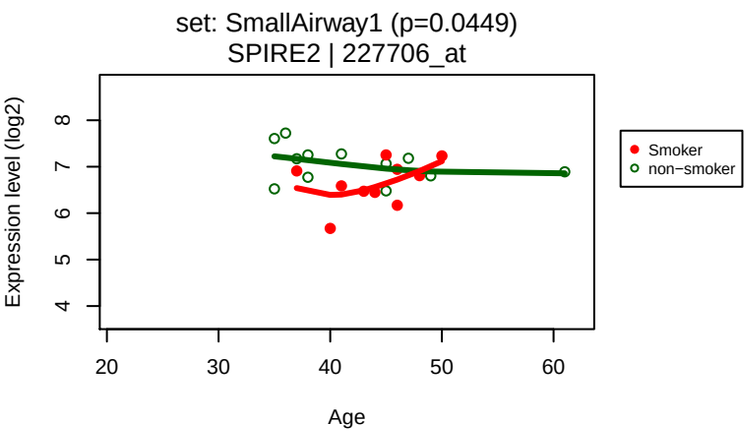
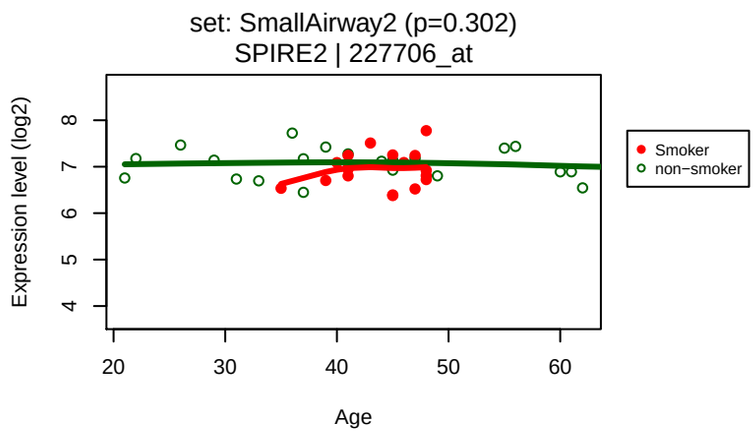
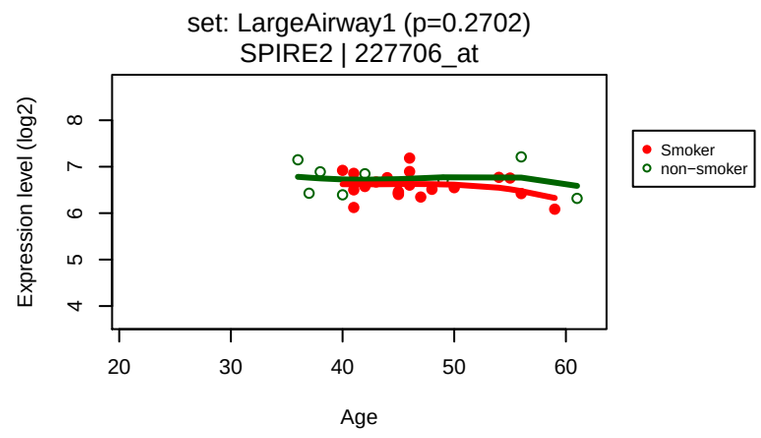

Gene expression of smokers and nonsmokers
gene: SPIRE2 | 227706_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0449
p=0.302
p=0.1837
p=0.2702
p=0.6977
p=0.4717
p=0.5788
p=0.6544
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

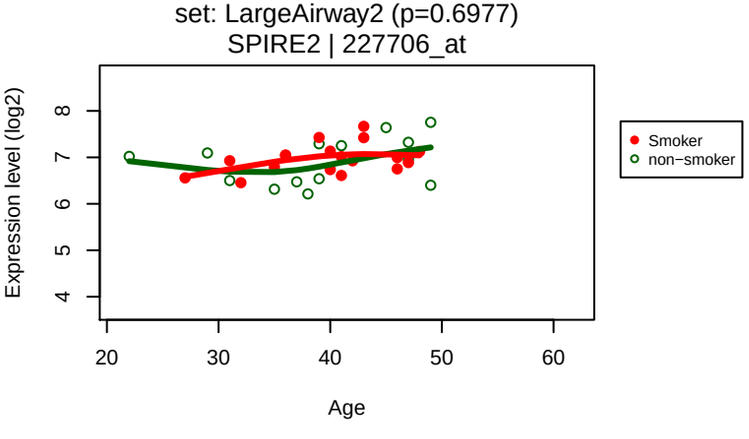

Gene expression ratio (smokers/nonsmokers)
gene: SPIRE2 | 227706_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0449
p=0.302
p=0.1837
p=0.2702
p=0.6977
p=0.4717
p=0.5788
p=0.6544
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

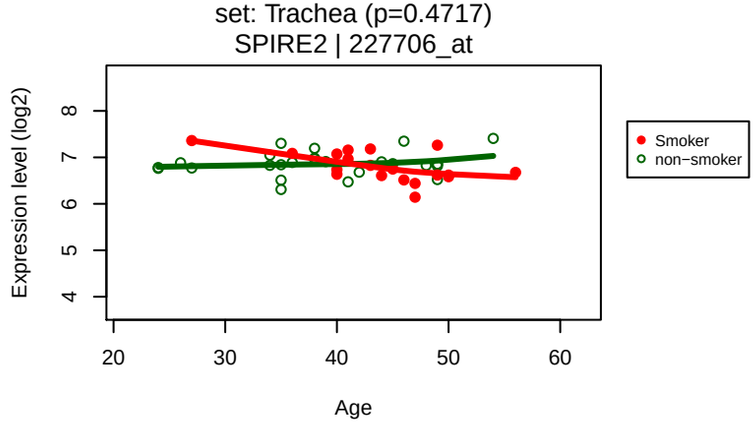

Gene expression ratio in Males
gene: SPIRE2 | 227706_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1143
p=0.0737
p=0.3504
p=0.9163
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPIRE2 | 227706_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5553
p=0.0221
p=0.6776
p=0.2293
SmallAirway2
LargeAirway2
Trachea
Macrophage2

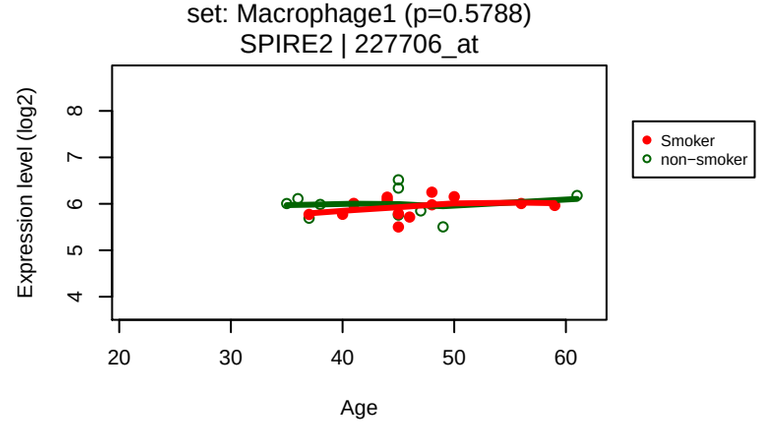

set: SmallAirway1 (p=0.0449)
SPIRE2 | 227706_at
Expression level (log2)
Age
Smoker
non−smoker

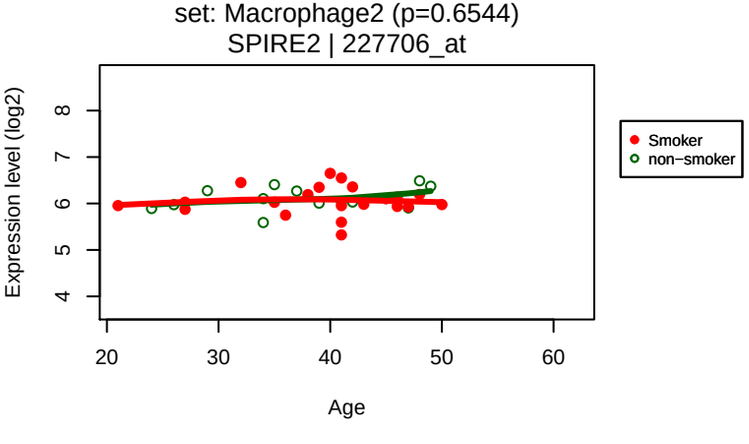

set: SmallAirway2 (p=0.302)
SPIRE2 | 227706_at
Expression level (log2)
Age
Smoker
non−smoker

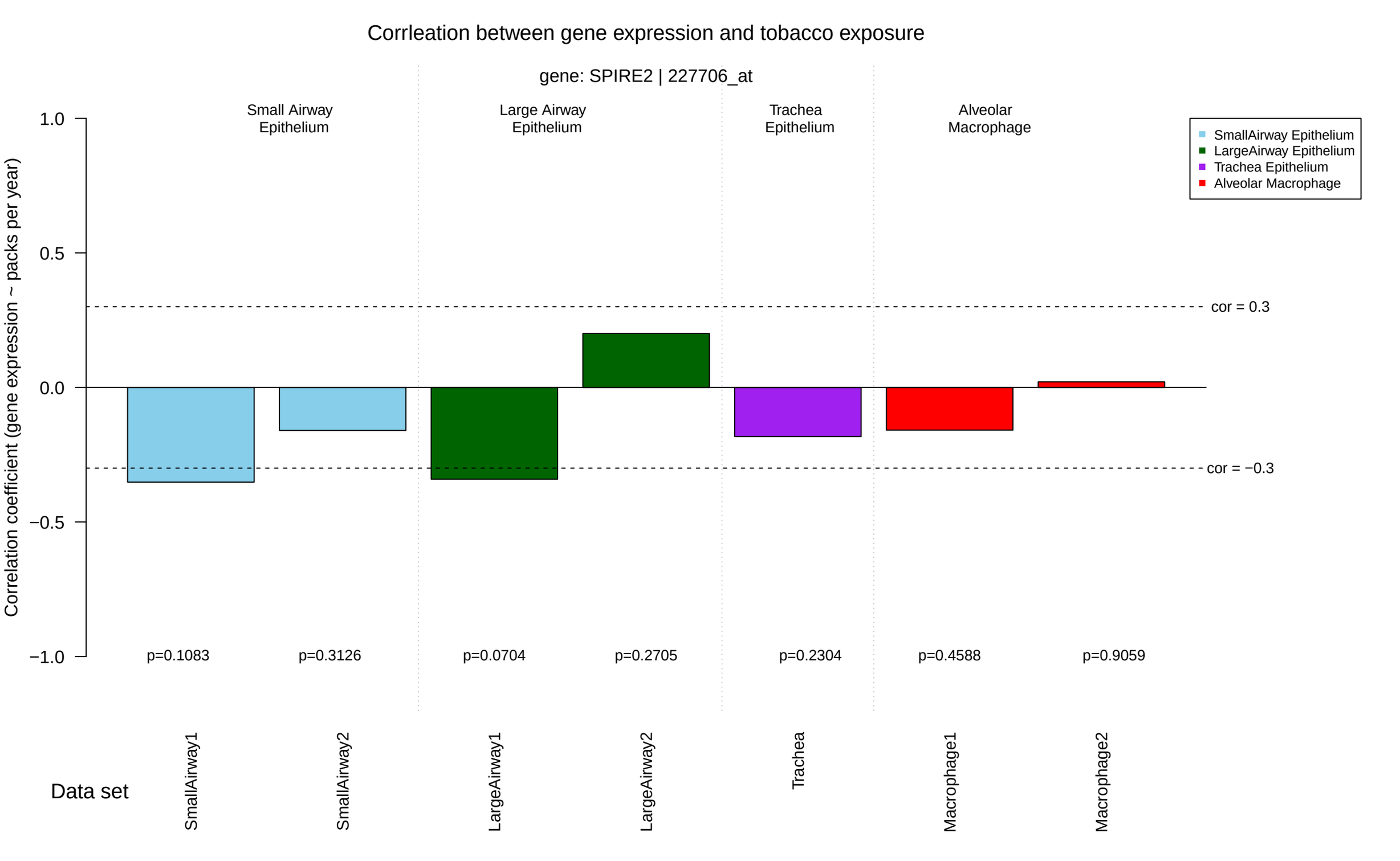

set: LargeAirway1 (p=0.2702)
SPIRE2 | 227706_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6977)
SPIRE2 | 227706_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4717)
SPIRE2 | 227706_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5788)
SPIRE2 | 227706_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6544)
SPIRE2 | 227706_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPIRE2 | 227706_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1083
p=0.3126
p=0.0704
p=0.2705
p=0.2304
p=0.4588
p=0.9059
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2